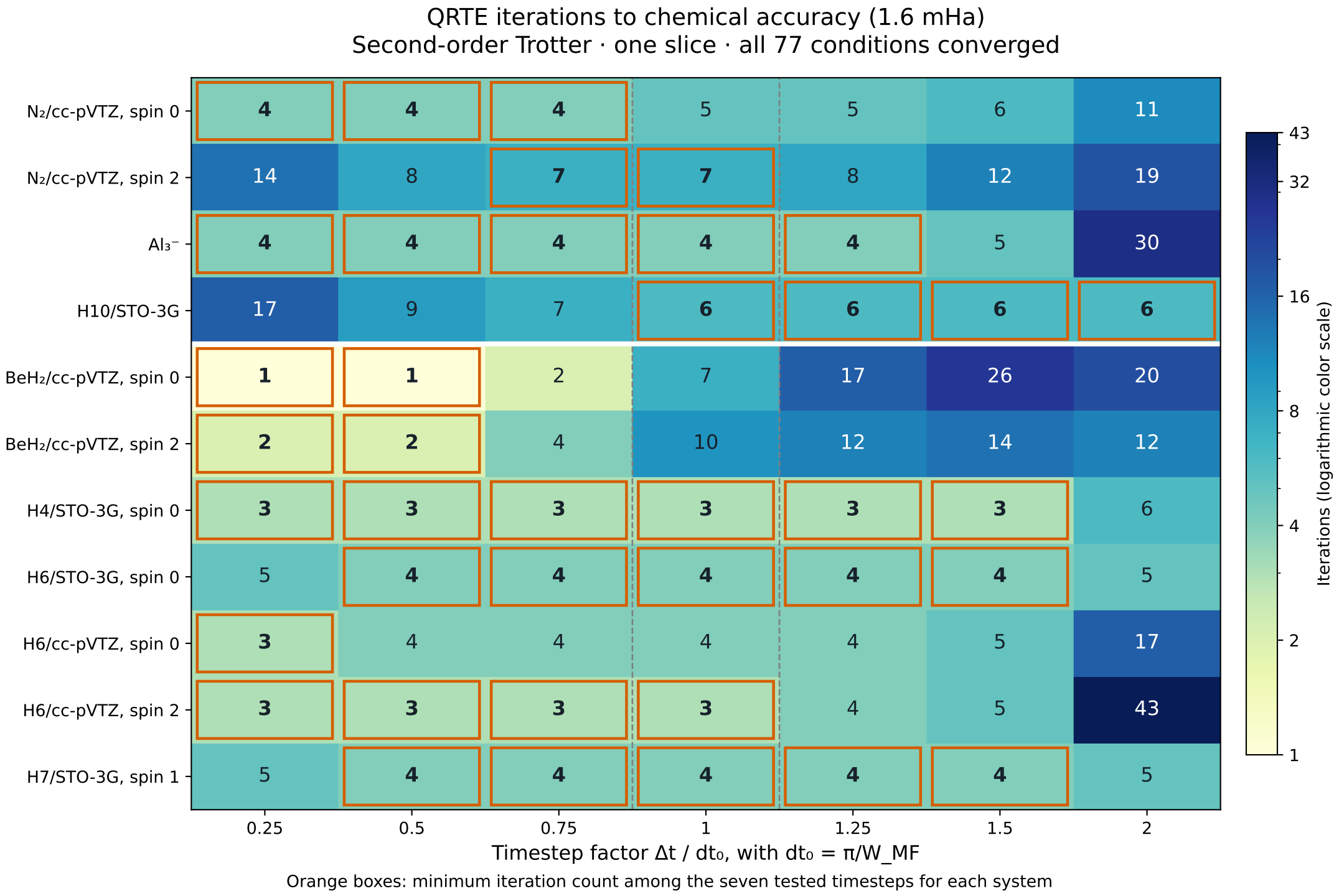
QRTE iterations to chemical accuracy (1.6 mHa)
Second-order Trotter · one slice · all 77 conditions converged
N₂/cc-pVTZ, spin 0
4 4 4 5 5 6 11
N₂/cc-pVTZ, spin 2
14 8 7 7 8 12 19
Al₃⁻
4 4 4 4 4 5 30
H10/STO-3G
17 9 7 6 6 6 6
BeH₂/cc-pVTZ, spin 0
1 1 2 7 17 26 20
BeH₂/cc-pVTZ, spin 2
2 2 4 10 12 14 12
H4/STO-3G, spin 0
3 3 3 3 3 3 6
H6/STO-3G, spin 0
5 4 4 4 4 4 5
H6/cc-pVTZ, spin 0
3 4 4 4 4 5 17
H6/cc-pVTZ, spin 2
3 3 3 3 4 5 43
H7/STO-3G, spin 1
5 4 4 4 4 4 5
0.25 0.5 0.75 1 1.25 1.5 2
Timestep factor Δt / dt₀, with dt₀ = π/W_MF
Iterations (logarithmic color scale)
1 2 4 8 16 32 43
Orange boxes: minimum iteration count among the seven tested timesteps for each system